lab-2-FGAI

Using Kaggle, Fast.AI, Hugging Face (Installation of Transformer Module, Sentiment analysis, NLP, Translation and summarization and sentence embedding)

```
# Install required libraries
!pip install --upgrade pip
!pip install transformers datasets sentence-transformers fastai[transformers] torch torchvision torchaudio kaggle

# Verify installs
import pkgutil
for pkg in ("transformers","datasets","sentence_transformers","fastai","kaggle","torch"):
    print(pkg, "installed:", pkgutil.find_loader(pkg) is not None)
```

```
Requirement already satisfied: pip in /usr/local/lib/python3.12/dist-packages (24.1.2)
Collecting pip
  Downloading pip-25.3-py3-none-any.whl.metadata (4.7 kB)
Downloading pip-25.3-py3-none-any.whl (1.8 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.8/1.8 MB 22.0 MB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 24.1.2
    Uninstalling pip-24.1.2:
      Successfully uninstalled pip-24.1.2
Successfully installed pip-25.3
WARNING: fastai 2.8.6 does not provide the extra 'transformers'
transformers installed: True
datasets installed: True
sentence_transformers installed: True
fastai installed: True
kaggle installed: True
torch installed: True
/tmp/ipython-input-3208982662.py:8: DeprecationWarning: 'pkgutil.find_loader' is deprecated and slated for remova
  print(pkg, "installed:", pkgutil.find_loader(pkg) is not None)
```

```
from transformers import pipeline

sentiment = pipeline("sentiment-analysis")

texts = [
    "I love this product, it works perfectly.",
    "This is the worst experience I have ever had.",
    "The movie was okay, not great but not terrible.",
    "Amazing performance and great story!",
    "Customer service was very disappointing."
] * 40
results = sentiment(texts)

results[:5]
```

```
WARNING:torchao.kernel.intmm:Warning: Detected no triton, on systems without Triton certain kernels will not work
No model was supplied, defaulted to distilbert/distilbert-base-uncased-finetuned-sst-2-english and revision 714eb
Using a pipeline without specifying a model name and revision in production is not recommended.
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/t
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
```

config.json: 100% 629/629 [00:00<00:00, 54.5kB/s]

model.safetensors: 100% 268M/268M [00:01<00:00, 215MB/s]

tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 4.59kB/s]

vocab.txt: 232k/? [00:00<00:00, 6.77MB/s]

```
Device set to use cpu
[{'label': 'POSITIVE', 'score': 0.9998784065246582},
 {'label': 'NEGATIVE', 'score': 0.9997627139091492},
 {'label': 'POSITIVE', 'score': 0.9776669144630432},
 {'label': 'POSITIVE', 'score': 0.999883770942688},
 {'label': 'NEGATIVE', 'score': 0.999774158000946}]
```

```
import csv

with open("sentiment_results.csv", "w", newline="", encoding="utf-8") as f:
```

```
    writer = csv.writer(f)
    writer.writerow(["text", "label", "score"])
    for t, r in zip(texts, results):
        writer.writerow([t, r["label"], r["score"]])

print("Saved sentiment_results.csv")
```

```
Saved sentiment_results.csv
```

```
from transformers import pipeline

summarizer = pipeline("summarization")

articles = [
    """Artificial intelligence has seen rapid growth in recent years.
    Large language models are now capable of understanding and generating text
    at near human levels. Researchers continue to explore ethical and safety concerns.""",

    """Climate change is impacting ecosystems worldwide. Rising temperatures,
    melting glaciers, and extreme weather events pose serious risks to biodiversity."""
] * 5

summaries = []

for article in articles:
    summary = summarizer(article, max_length=20, min_length=10, do_sample=False)[0]["summary_text"]
    summaries.append(summary)

summaries[:3]
```

```
No model was supplied, defaulted to sshleifer/distilbart-cnn-12-6 and revision a4f8f3e (https://huggingface.co/ss
Using a pipeline without specifying a model name and revision in production is not recommended.
Device set to use cpu
[' Artificial intelligence has seen rapid growth in recent years . Large language models are now capable of',
 ' Climate change is impacting ecosystems worldwide . Rising temperatures, melting glaciers, and extreme weather
events',
 ' Artificial intelligence has seen rapid growth in recent years . Large language models are now capable of']
```

```
from transformers import pipeline

en_fr = pipeline("translation_en_to_fr", model="Helsinki-NLP/opus-mt-en-fr")
fr_en = pipeline("translation_fr_to_en", model="Helsinki-NLP/opus-mt-fr-en")

sentences = [
    "Machine learning is changing the world.",
    "Artificial intelligence is fascinating.",
    "This course is very informative."
] * 7   # 21 sentences

translations = []

for s in sentences:
    fr = en_fr(s)[0]["translation_text"]
    back = fr_en(fr)[0]["translation_text"]
    translations.append((s, fr, back))

translations[:3]
```

```
config.json:        1.42k/? [00:00<00:00, 163kB/s]

pytorch_model.bin: 100%                                                    301M/301M [00:06<00:00, 28.2MB/s]

generation_config.json: 100%                                                   293/293 [00:00<00:00, 35.7kB/s]

tokenizer_config.json: 100%                                                  42.0/42.0 [00:00<00:00, 4.85kB/s]

model.safetensors: 100%                                                    301M/301M [00:02<00:00, 155MB/s]

source.spm: 100%                                                         778k/778k [00:00<00:00, 54.8MB/s]

target.spm: 100%                                                        802k/802k [00:00<00:00, 45.7MB/s]

vocab.json:        1.34M/? [00:00<00:00, 61.9MB/s]
/usr/local/lib/python3.12/dist-packages/transformers/models/marian/tokenization_marian.py:175: UserWarning: Recom
  warnings.warn("Recommended: pip install sacremoses.")
Device set to use cpu
config.json:        1.42k/? [00:00<00:00, 68.8kB/s]

model.safetensors: 100%                                                    301M/301M [00:02<00:00, 241MB/s]

generation_config.json: 100%                                                   293/293 [00:00<00:00, 5.10kB/s]

tokenizer_config.json: 100%                                                  42.0/42.0 [00:00<00:00, 1.41kB/s]

source.spm: 100%                                                         802k/802k [00:00<00:00, 14.4MB/s]

target.spm: 100%                                                        778k/778k [00:00<00:00, 17.1MB/s]

vocab.json:        1.34M/? [00:00<00:00, 6.06MB/s]
Device set to use cpu
[('Machine learning is changing the world.',
  "L'apprentissage automatique change le monde.",
  'Machine learning is changing the world.'),
 ('Artificial intelligence is fascinating.',
  "L'intelligence artificielle est fascinante.",
  'Artificial intelligence is fascinating.'),
 ('This course is very informative.',
  'Ce cours est très instructif.',
  'This course is very instructive.')]
```

```
from sentence_transformers import SentenceTransformer, util

model = SentenceTransformer("all-MiniLM-L6-v2")

faqs = [f"How do I reset my password question {i}?" for i in range(100)]
faq_embeddings = model.encode(faqs, convert_to_tensor=True)

query = "How can I change my password?"
query_embedding = model.encode(query, convert_to_tensor=True)

scores = util.pytorch_cos_sim(query_embedding, faq_embeddings)[0]

top_results = scores.topk(3)

for idx in top_results.indices:
    print(faqs[idx])
```

```
modules.json: 100%                                                       349/349 [00:00<00:00, 39.1kB/s]

config_sentence_transformers.json: 100%                                                  116/116 [00:00<00:00, 14.2kB/s]

README.md:        10.5k/? [00:00<00:00, 965kB/s]

sentence_bert_config.json: 100%                                                   53.0/53.0 [00:00<00:00, 5.91kB/s]

config.json: 100%                                                       612/612 [00:00<00:00, 58.8kB/s]

model.safetensors: 100%                                                    90.9M/90.9M [00:01<00:00, 82.8MB/s]

tokenizer_config.json: 100%                                                  350/350 [00:00<00:00, 43.7kB/s]

vocab.txt:        232k/? [00:00<00:00, 15.7MB/s]

tokenizer.json:        466k/? [00:00<00:00, 21.6MB/s]

special_tokens_map.json: 100%                                                   112/112 [00:00<00:00, 14.8kB/s]

config.json: 100%                                                       190/190 [00:00<00:00, 24.0kB/s]
How do I reset my password question 3?
How do I reset my password question 2?
How do I reset my password question 4?
```

```
# ==============================
# FastAI IMDb Text Classification
```

```
# Progress-bar safe for Colab
# ==============================

# ---- Fix fastprogress / Colab bug ----
import fastprogress.fastprogress as fpd
fpd.force_console_is_str = True
fpd.NO_BAR = True

from fastai.callback.progress import ProgressCallback

# ---- Imports ----
from datasets import load_dataset
import pandas as pd
from fastai.text.all import *

# ---- Load IMDb dataset ----
ds = load_dataset("imdb")

# Use smaller subset (lab-friendly)
train_ds = ds["train"].shuffle(seed=42).select(range(4000))
test_ds  = ds["test"].shuffle(seed=42).select(range(2000))

# Convert to DataFrame
train_df = pd.DataFrame(train_ds)
test_df  = pd.DataFrame(test_ds)

# ---- Create DataLoaders ----
dls = TextDataLoaders.from_df(
    train_df,
    text_col="text",
    label_col="label",
    valid_pct=0.1,
    bs=32
)

# ---- Create learner WITHOUT progress bars ----
learn = text_classifier_learner(
    dls,
    AWD_LSTM,
    metrics=accuracy,
    cbs=[]
)

# Remove any automatically added progress callbacks
learn.remove_cbs(ProgressCallback)

# ---- Train model (1 epoch) ----
learn.fine_tune(1)

# ---- Validate & print accuracy ----
val_loss, val_acc = learn.validate()
print(f"Validation Accuracy: {val_acc:.4f}")
```

```
---------------------------------------------------------------------------
KeyboardInterrupt                         Traceback (most recent call last)
/tmp/ipython-input-2807304535.py in <cell line: 0>()
     48
     49 # ---- Train model (1 epoch) ----
---> 50 learn.fine_tune(1)
     51
     52 # ---- Validate & print accuracy ----

                              27 frames
/usr/local/lib/python3.12/dist-packages/torch/nn/modules/rnn.py in forward(self, input, hx)
   1125
   1126         if batch_sizes is None:
-> 1127             result = _VF.lstm(
   1128                 input,
   1129                 hx,

KeyboardInterrupt:
```

```
from transformers import pipeline

# Create translation pipelines
en_to_fr = pipeline("translation_en_to_fr", model="Helsinki-NLP/opus-mt-en-fr")
fr_to_en = pipeline("translation_fr_to_en", model="Helsinki-NLP/opus-mt-fr-en")

# Sentences to translate
sentences = [
    "Machine learning is changing the world.",
```

```
    "Artificial intelligence is used in many applications.",
    "This lab is very helpful for understanding NLP."
]

print("ENGLISH → FRENCH → ENGLISH\n")

for s in sentences:
    french = en_to_fr(s)[0]["translation_text"]
    back_to_english = fr_to_en(french)[0]["translation_text"]

    print("Original English :", s)
    print("French           :", french)
    print("Back to English  :", back_to_english)
    print("-" * 50)
```

```
/usr/local/lib/python3.12/dist-packages/transformers/models/marian/tokenization_marian.py:175: UserWarning: Recom
  warnings.warn("Recommended: pip install sacremoses.")
Device set to use cpu
Device set to use cpu
ENGLISH → FRENCH → ENGLISH

Original English : Machine learning is changing the world.
French           : L'apprentissage automatique change le monde.
Back to English  : Machine learning is changing the world.
--------------------------------------------------
Original English : Artificial intelligence is used in many applications.
French           : L'intelligence artificielle est utilisée dans de nombreuses applications.
Back to English  : Artificial intelligence is used in many applications.
--------------------------------------------------
Original English : This lab is very helpful for understanding NLP.
French           : Ce laboratoire est très utile pour comprendre le NLP.
Back to English  : This laboratory is very useful for understanding NLP.
--------------------------------------------------
```

```
from google.colab import drive
drive.mount("/content/drive")

!cp translation_results_en_fr_en.csv /content/drive/MyDrive/
print("File copied to Google Drive")
```

```
Mounted at /content/drive
cp: cannot stat 'translation_results_en_fr_en.csv': No such file or directory
File copied to Google Drive
```